Review for T-RO Submission ID 25-1489

1 Main paper

This paper pertains to a framework for motion planning, the crux of which is solving mixed-integer convex programs (MICPs). This MICP formulation allows the online planner/controller to plan ahead footstep sequences *while remaining physically consistent* – which is an important feature to have for locomotion across uneven terrains (stairs, but also across platforms and even rebar, an interesting example showcased in the paper).

The paper is an extension of an IROS 2025 paper, which is reference [24] in the manuscript. As in [24], due to how expensive MICPs are to solve, the authors also rely on a "symbolic repair" mechanism and simplifications to the MICP when doing online replanning.

Main comments

The introduction motivates the problem very well, with an emphasis on safety-critical scenarios which contrasts with the lack of guarantees in mainstream heuristic-based approaches for footstep planning. The mixed-integer formulation itself is motivated by two related things in locomotion footstep planning: selecting the contact mode for each limb (hence the timing) and selecting the contact *surface*.

However, where the introduction falters is that it doesn't really, clearly define what the authors mean by "reactive synthesis" while employing it several times, along with "LTL". This is actually from the subfield of logic in (pure-r) computer science. The reader may try to parse what the authors mean by "synthesis" in the second paragraph of p.2 which gives a brief description of the planning framework – that it would be the idea of generating, offline, a set of locomotion skills using the *gait-free* (i.e. gait is part of the optimization) MICP, and also "stitching" these skills together online through the *gait-fixed* MICP. But then, it would sound like some kind of physics-aware TAMP or otherwise said contact-rich trajectory optimization with simultaneous task planning? Then, the reader would ask what "*reactive*" synthesis is. Would it mean "real-time"? Environment-aware? I do not know if this is all standard in the TAMP-related literature. Furthermore, the introduction early on talks of "LTL-based reactive synthesis" without the reader necessarily knowing what LTL is either.

In fact, the reader could guess by actually looking at the content of reference [23], "Synthesis of Reactive(1) Designs" by Piterman et al, which goes over LTL formulas, synthesis, and so on, or wait until §III to finally understand that this is about formal logic. My point here is that the robotics community encompasses a diverse set of technical backgrounds, and in a (long) T-RO paper the authors have more room for hand-holding the reader and being a bit more educational. This also means that the order in which concepts are introduced and described is important for clarity. In the introduction, the authors could more strongly direct the reader to read [23] before continuing with the rest of the paper, or direct them towards the reference [98] cited in §III from the start, because by then the reader could be frustrated/confused. The authors could, in the introduction still, even quickly describe what LTL is as they would to a layperson, and motivate its use for this problem (i.e. why it is appropriate, perhaps even provide a comparison to very popular approaches familiar to ML people such as MCTS).

The related work §II generally gives a good, structured overview of approaches in the literature around terrain-adaptive locomotion, with a progression from hierarchical planning (sometimes with heuristics) to simultaneous (mixed-integer optimization) planning and TAMP, before finally introducing "physically-feasible reactive synthesis". However, the related work section still suffers from the problem of the introduction, using the term "synthesis" and "Linear Temporal Logic (LTL)" as if the reader already grappled with these concepts. Another criticism I have of the related work section (§II) is that there is no mention at all of learning-based approaches (e.g. reinforcement learning), although these have become incredibly widespread in both the literature and industrial[1] applications.

Main strengths

The paper adds some important content compared to the main point of reference, the IROS paper [24].

[1]Meaning at least demonstrations, publications and (marketing) materials from robot makers e.g. Unitree.

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First is of course the extensive (including real-world) experiments as shown in the companion video. They provide a rather strong argument for the method's versatility.

These experiments also include a benchmark in simulation, against pure MIP planning (§ IX.E). This is a good addition to the paper to give the reader an idea of how a "pure" planner (without the proposed ingredients: symbolic planning, repair mechanism...) would work, and the visualization in Fig. 9 is great to give a bird's eye view of each benchmarked scenario.

Likely stemming from the need for real hardware experiments, the paper also adds a tracking control module based on nonlinear MPC (in §VIII).

Main criticisms

The authors do not mention a couple implementation and runtime details which I would deem very important:

1. Which solver is used to solve the mixed-integer (convex) program? Is it Gurobi? CPLEX? Some other, homegrown MI(C)P solver? Have the authors compared how different solvers would perform? In contrast the authors *do* mention that they are using OCS2 as a nonlinear MPC framework.

2. What kind of hardware (outside of the robot – computer CPU, RAM) are the computations running on? AMD? Intel? Which architecture? How many cores?

Beyond the introduction, the rest of the paper (as well as [24]) uses the term "reactive synthesis" 14 times (outside of the title itself) without really, properly defining what those terms mean together. This is one of my main problems with the paper in terms of clarity for readers unfamiliar with the formal logic aspect, as I've noted above. This is not an easy read.

I think unfortunately the comparison with other methods is a tad lacking. The authors compare with a heuristic on real hardware. What about another more sophisticated approach? Another hierarchical planner? A learning-based method?

Minor comments

Typos:

- *bottom of p.2 in §II.B:* "high~~ly~~ nonlinearity"

- *p.3:* "and timing are chosen *a priori*"

As in [24], this paper focuses on quadrupedal walking. Does this framework extend to bipedal robots (with some adaptation of the formulation for the arms)? If the authors could provide a quick answer or comment on it, I think the conclusion would be appropriate for that (as part providing a perspective on future work in the area, broadening the paper's reach in a way).

Sections IV and V are very short. This is the most minor of minor comments but I feel the authors could rebalance a bit by merging them into a single section – "Problem statement and approach"? Or could it be a subsection of §III?

Fig. 9, for the simulation benchmark (from §IX.E), is called a "histogram". It's rather a bar chart of the averages. I would like to see error bars/whiskers on here, to see the standard deviations. Furthermore, a repeat of my previous point, knowing the actual underlying MIP solver would be a good indication to the reader.

Table IV (solve times for the online *gait-fixed* MICP) would benefit from including the standard deviation for the solve times (to see how concentrated around the mean values they are). Or being supported/replaced by a histogram so we could get a clear idea of the distribution of solve times.

2 Multimedia attachment

The paper comes with a nearly 4-minute video explaining the workflow behind the proposed motion planning framework and showcasing it running on a quadrupedal robot (Unitree Go2/SkyMul Chotu) both in simulation and (a contribution of this paper compared to the one which introduced the framework in [24]) on real hardware.

The shown motion plans are I would say somewhat impressive for something entirely model-based. Showing how the heuristic footstep planner does (and fails) is also a good comparison point.

The (generated?) voice-over explanations are a nice touch which keep the video itself uncluttered. They are quite clear.

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